



How much is rain worth? Nevada lawmakers weigh continued funding for Desert Research Institute's cloud seeding program



Pahrump ground-based cloud seeding generator looking East towards Mount Charleston Photo by: Jesse Juchtzer / DRI /Courtesy

By [Ilana Williams \(contact\)](#)

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In a state where every raindrop counts, scientists in the Desert Research Institute's Cloud Seeding Program have mastered the art of enhancing natural precipitation, transforming wispy clouds into meaningful rainfall and snowpack across Nevada's parched landscape.

Lawmakers are weighing a measure that would lock in funding for the research through 2027.

Last session, lawmakers approved Senate Bill 99 [to allocate \\$1.2 million](#) to the program from 2023 to 2025. The current proposal, Senate Bill 6, would maintain funding levels for the 2025-2027 biennium.

Frank McDonough, director of the institute's cloud seeding program, said previous funding enabled repairs to 25 cloud seeding generators and weather stations across the state.

These investments yielded significant results: Last winter alone, DRI estimated an additional 56,000 acre-feet of snowmelt was added to four key Nevada watersheds — the Humboldt River Basin, Ruby Mountains, Truckee River Basin and Spring Mountains.

The institute wants the cloud seeding program to be part of a long-term management approach to Nevada's water supply and water resources, McDonough said.

"The cloud seeding seems to be able to partially put the clouds back to the efficiency that they probably would have had prior to human air pollution participating in the precipitation processes now," McDonough said.

The four cloud seeding generators positioned at the base of the Spring Mountains northwest of Las Vegas generated 14,000 acre-feet of water — enough to supply the annual water needs of approximately 42,000 households, McDonough said.

The cloud seeding process operates from ground-based generators during widespread storm systems with low cloud coverage. This strategic approach can enhance precipitation by up to 10% across the targeted basins, he said.

"We can enhance our snowfall, increase our snowpack and increase the amount of water in our rivers, streams and lakes," said state Sen. Melanie Scheible, D-Las Vegas. "We can't create the storms but when we do have storms, we can help make the storm clouds more efficient."

The proposal was heard last month by the Senate Committee on Natural Resources. No vote was taken.

Tracy Bower, director of external affairs at Desert Research Institute, said the program has a long history, operating continuously since 1976 until budget cuts during the Great Recession forced its suspension in 2009.

In the interim, it collaborated with local water agencies to maintain cloud seeding operations when possible, though these fragmented efforts proved less efficient and more costly than a coordinated statewide program.

Bower noted that local sponsors typically only fund cloud seeding when drought conditions have already reached critical levels for water supplies.

McDonough emphasized that while annual snowpack levels naturally fluctuate, rising temperatures have intensified water demands by accelerating evaporation rates. He explained that high-elevation snowpack historically persisted until June or July, providing a steady water supply throughout early summer.

Now, these crucial water reserves are depleting faster, highlighting the growing importance of enhanced precipitation technologies in [Nevada's changing climate](#).

“What we’re really seeing is the snow is gone earlier,” McDonough said. “We’re down to bare earth three weeks earlier than we historically have been.”

Meteorologists identify clouds containing subfreezing liquid water droplets before activating generators that release silver iodide particles into air currents entering these clouds, he explained.

These microscopic particles provide a surface for water droplets to freeze upon, creating ice crystals in clouds that naturally lack them. The crystals grow rapidly, forming ice and large snowflakes that eventually fall as precipitation.

He stressed that silver iodide has been extensively studied for over 60 years and proven non-hazardous to the environment. While silver iodide occurs naturally in the atmosphere at parts per trillion, the cloud seeding process releases concentrations a million times lower than these naturally occurring levels.

“If we cloud seed for a million years, you might see one more particle in the soil than you’re seeing now,” McDonough said.

He added that during a typical winter season, Desert Research Institute targets approximately six storm systems for cloud seeding operations. The entire network of generators is remotely operated, with a sophisticated weather station and camera system positioned between Kyle Canyon and Lee Canyon at an elevation of 11,000 feet.

This high-altitude monitoring equipment allows technicians to observe surfaces where subfreezing water droplets form, enabling them to precisely identify which cloud formations are most suitable for the seeding process.

“It’s a tool in the toolbox our state can utilize to help with the impacts of climate change and our water supply,” Scheible said.